

Tim Gallagher

Rectangular Cantilever + x Comp. Stress

Static structural comparison

Geometry

$$L = 6 \text{ mm}$$

$$= 0.006 \text{ m}$$

$$W = 2 \text{ mm}$$

$$= 0.002 \text{ m}$$

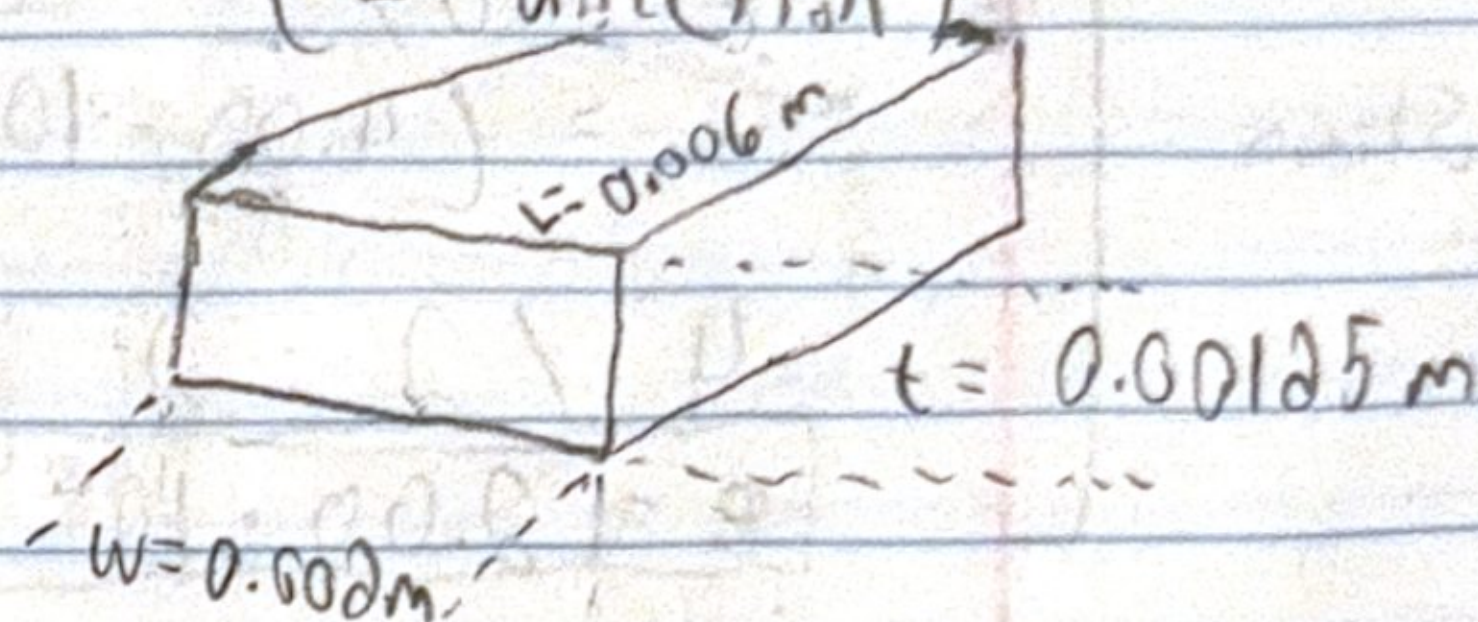
$$t = 1.25 \text{ mm}$$

$$= 0.00125 \text{ m}$$

(x direction)

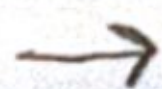
(y direction)

(z direction)



Load

$$F_x = +100 \text{ N}$$



Material

Structural Steel ($E = 2 \cdot 10^{11} \text{ Pa}$)

$$1 \text{ GPa} = 10^9 \text{ Pa}$$

$$E = 200 \cdot 10^9 \text{ Pa}$$

$$E \approx 2 \cdot 10^{11} \text{ Pa}$$

$$\approx 200 \text{ GPa}$$

$$\nu = 0.30$$

Poisson's

$$\nu = -\frac{\epsilon_{lat}}{\epsilon_{axial}}$$

ϵ_{axial}

1) Cross Sectional Area

$$A = wt$$

$$= (0.002 \text{ m}) (0.00125 \text{ m})$$

$$= 0.0000025 \text{ m}^2$$

$$= 2.50 \cdot 10^{-6} \text{ m}^2$$

Loaded Face Area

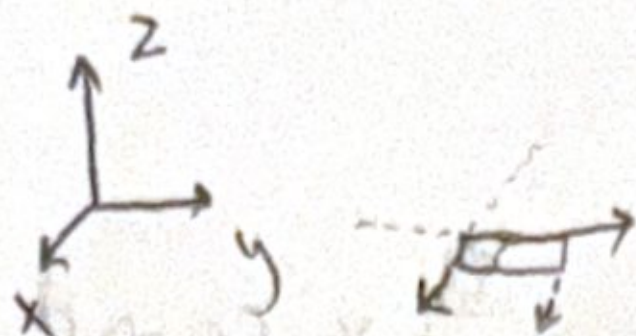


$$\sigma = F/A$$

$$= 100 \text{ N} / (2.50 \cdot 10^{-6} \text{ m}^2)$$

$$= 40,000,000 \text{ N/m}^2$$

Avg. Stress



$$1 \text{ Pa} = 1 \text{ N/m}^2$$

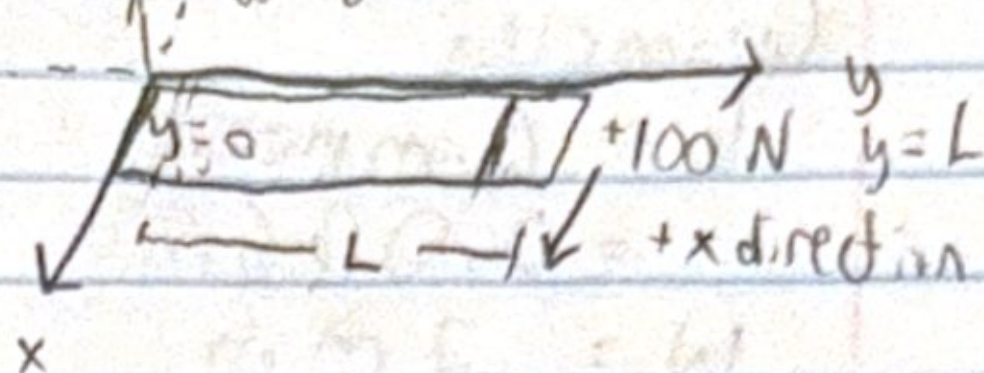
$$\sigma = 4.00 \cdot 10^7 \text{ Pa}$$

$$\sigma = 40 \text{ MPa}$$

End face Area
 $A = wt$

$$t = 1.25 \text{ mm}$$

$$w = 2 \text{ mm}$$



Strain

$$E = \sigma / \epsilon$$

$$\epsilon = \sigma / E$$

$$= (4.00 \cdot 10^7 \text{ Pa}) / (2.00 \cdot 10^{11} \text{ Pa})$$

Young's modulus

$$4 / 2 = 2 \text{ and } 10^7 / 10^{11} = 10^{-4}$$

$$\epsilon = 2.00 \cdot 10^{-4}$$

$$\nu = - \frac{\epsilon_{\text{lateral}}}{\epsilon_{\text{axial}}}$$

$$\epsilon_{\text{lateral}} = - (\nu) (\epsilon_{\text{axial}})$$

$$= - (0.30) (2.00 \cdot 10^{-4})$$

$$= -6.00 \cdot 10^{-5}$$

Young's

$$E = \sigma / \epsilon$$

$$= (4.00 \cdot 10^7) / (2.00 \cdot 10^{-4})$$

$$= 2.00 \cdot 10^{11} \text{ Pa} \quad \checkmark$$

$$= 200 \text{ GPa} \quad \checkmark$$

2)

Avg Applied Traction (Avg. Stress)

$$p = F/A$$

$$p = 100 \text{ N} / (2.50 \cdot 10^{-6} \text{ m}^2)$$

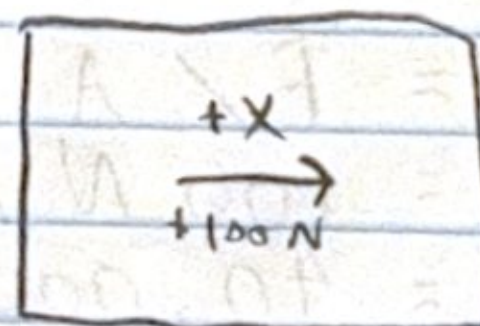
$$p = 40,000,000 \text{ N/m}^2$$

$$\approx 4.00 \cdot 10^7 \text{ Pa}$$

$$1 \text{ MPa} = 10^6 \text{ Pa}$$

$$p = 40 \text{ MPa}$$

y=L



3) Cantilever Bending Moment

$$y = 0 \text{ (fixed)}$$

$$y = L \text{ (applied) } \quad \text{*length}$$

loaded face

$$M_{\max} = FL$$

$$M_{\max} = (100 \text{ N})(0.006 \text{ m})$$

$$M_{\max} = \boxed{0.6 \text{ N}\cdot\text{m}}$$

4) Area Moment of Inertia

$$I_z = \frac{tw^3}{12}$$

$$= \frac{(0.00125)(0.002)^3}{12}$$

$$= \boxed{8.33 \cdot 10^{-13} \text{ m}^4}$$

5) Maximum Bending Stress

$$\sigma_b = \frac{Mc}{I}$$

* M is bending stress

* c is distance from neutral axis to outer surface

$$M = 0.600 \text{ N}\cdot\text{m}$$

$$I_z = 8.33 \cdot 10^{-13} \text{ m}^4$$

* I is area moment of inertia

$$c = w/2$$

$$= (0.002) / 2$$

$$= 0.001 \text{ m}$$

$$\sigma_{b \max} = \frac{M_{\max} c}{I_z}$$

$$= \frac{(0.600)(0.001)}{8.33 \cdot 10^{-13}}$$

$$= 7.20 \cdot 10^8 \text{ Pa}$$

$$1 \text{ MPa} = 10^6 \text{ Pa}$$

$$\sigma_{b \max} = \boxed{720 \text{ MPa}}$$

6) Tip Deflection

$$\delta_{max} = \frac{FL^3}{3EI_z}$$

$$F_x = 100 \text{ N}$$

$$L = 0.006 \text{ m}$$

$$E = 2.00 \cdot 10^{11} \text{ Pa}$$

$$I_z = 8.33 \cdot 10^{-13} \text{ m}^4$$

$$\delta_x = \frac{(F_x)L^3}{(3)E(I_z)}$$

$$= \frac{(100) (0.006)^3}{3 (2.00 \cdot 10^{11}) (8.33 \cdot 10^{-13})}$$

$$= 2.16 \cdot 10^{-5}$$

$$3 (1.667 \cdot 10^{-1})$$

$$= 4.32 \cdot 10^{-5} \text{ m}$$

$$= \boxed{0.0432 \text{ mm}}$$

deformation

7) Equivalent Stress (von Mises)

$$\sigma_{vm} = \sqrt{\frac{(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2}{2} + 6\tau_{xy}^2 + 6\tau_{yz}^2 + 6\tau_{zx}^2}$$

* one directional bend

$$\sigma_{vm} \approx |\sigma_x|$$

$$\approx \boxed{720 \text{ MPa}}$$